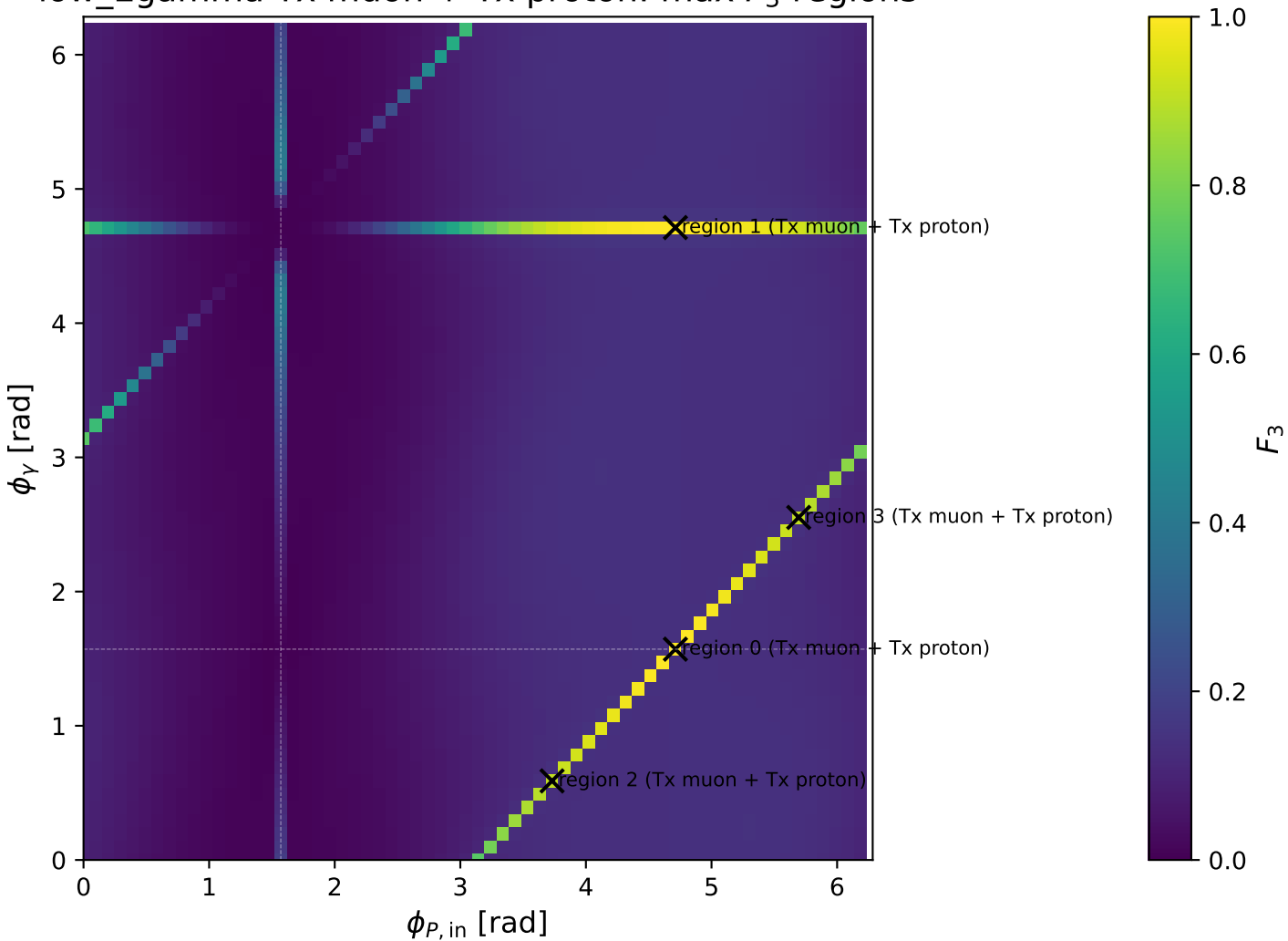
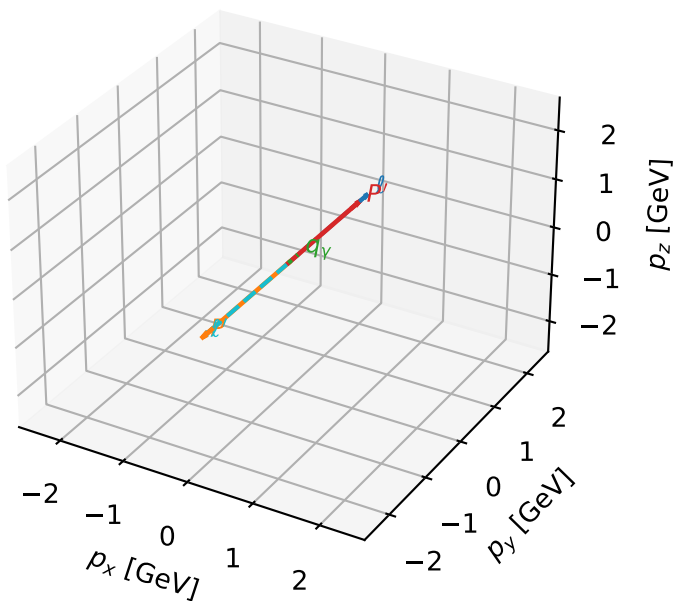


low_Egamma Tx muon + Tx proton: max F_3 regions



Tx muon + Tx proton characteristic max F_3 configuration

3D momenta

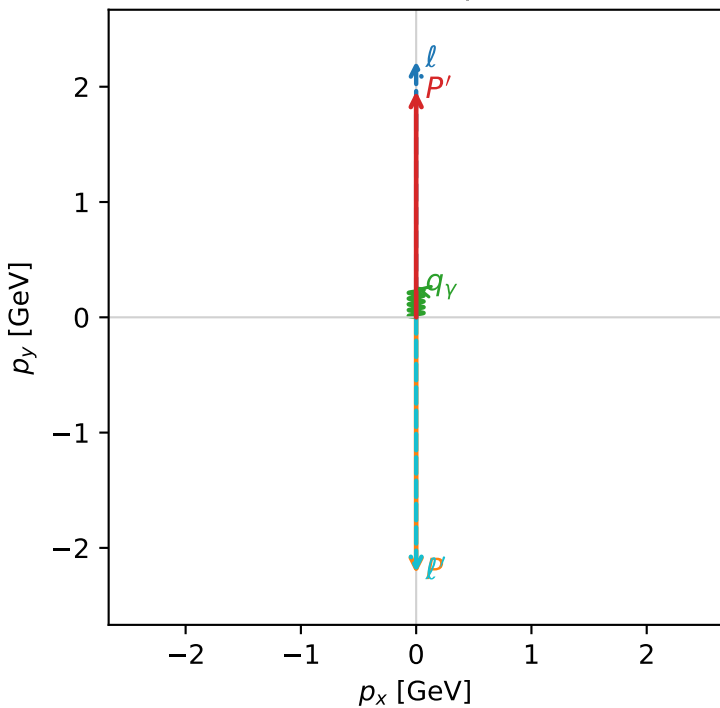


f3_low_egamma_txtx_region_0 F_3=1
 region: low_Egamma_TxTx_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.96163$
 $\theta_{in}=1.57079$, $\phi_l=1.5708$, $\phi_p=4.71239$
 $E_Y=0.25$, $\phi_Y=1.5708$
 $Q^2=19.6668$, $x_B=0.994621$, $t=-17.4552$, $W^2=0.986208$, $y=0.958708$
 $\theta(l', \gamma)=180$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= 0.0000$
 ℓ' $E= 2.2142$ $p_x= -0.0000$ $p_y= -2.2116$ $p_z= 0.0000$
 P' $E= 2.1744$ $p_x= 0.0000$ $p_y= 1.9616$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= 0.0000$ $p_y= 0.2500$ $p_z= 0.0000$

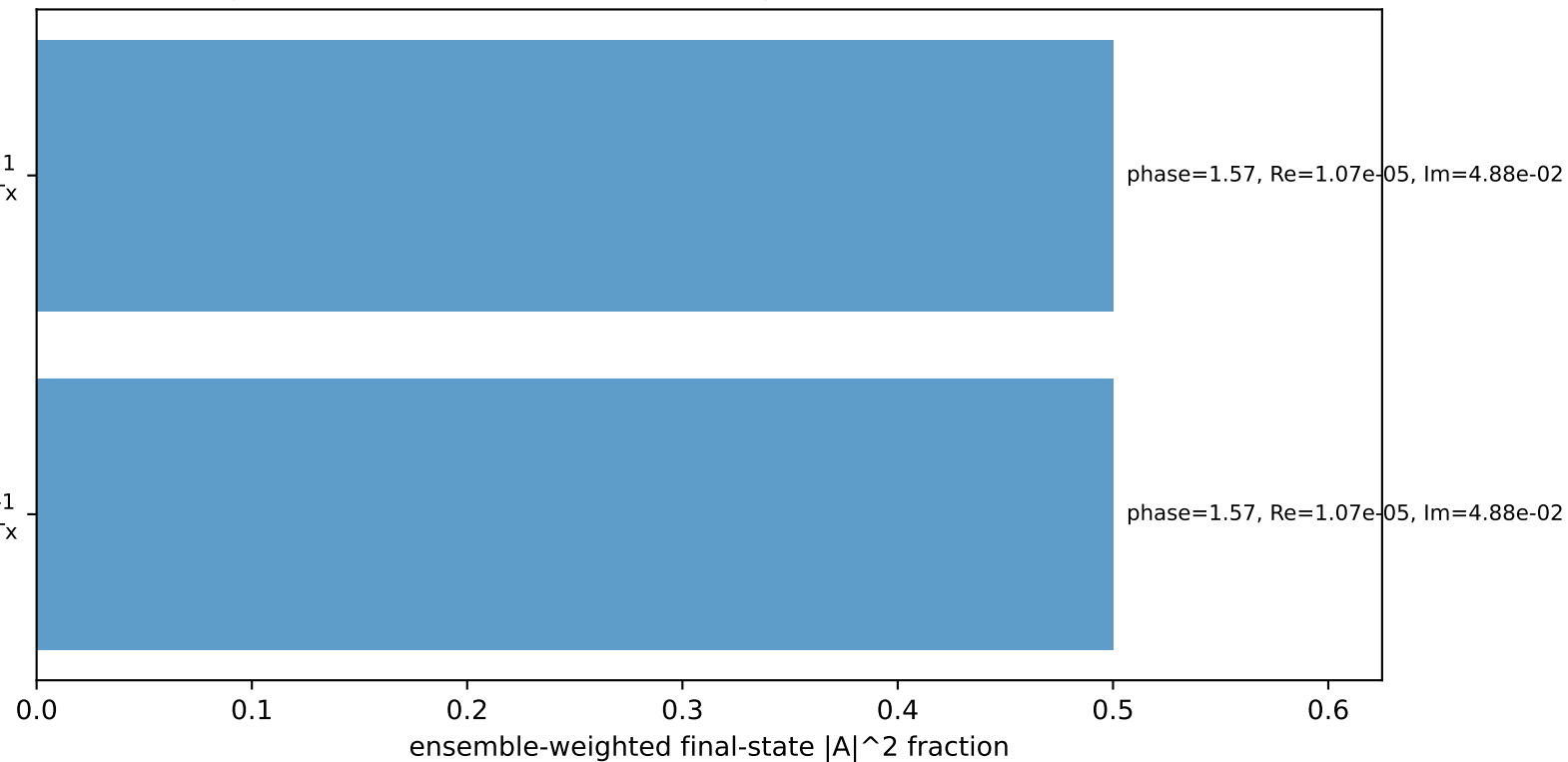
Transverse plane



$h_e=-1$, $h_p=-1$, $h_Y=+1$
 electron Tx, proton Tx

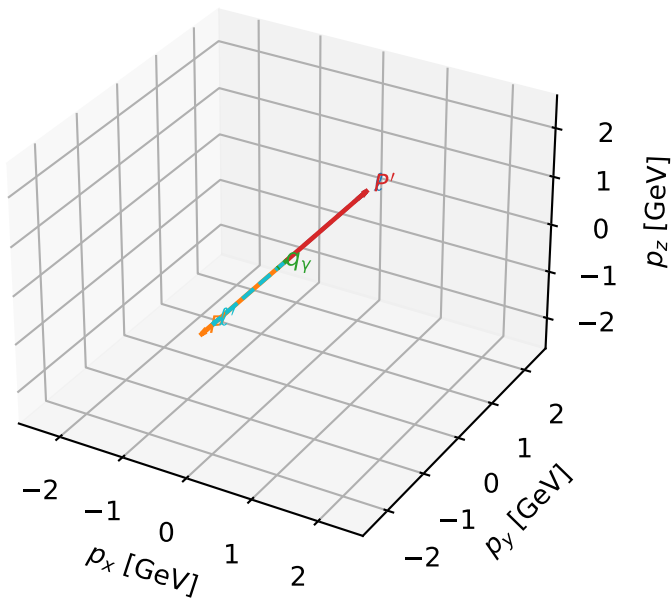
$h_e=+1$, $h_p=+1$, $h_Y=-1$
 electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



Tx muon + Tx proton characteristic max F_3 configuration

3D momenta

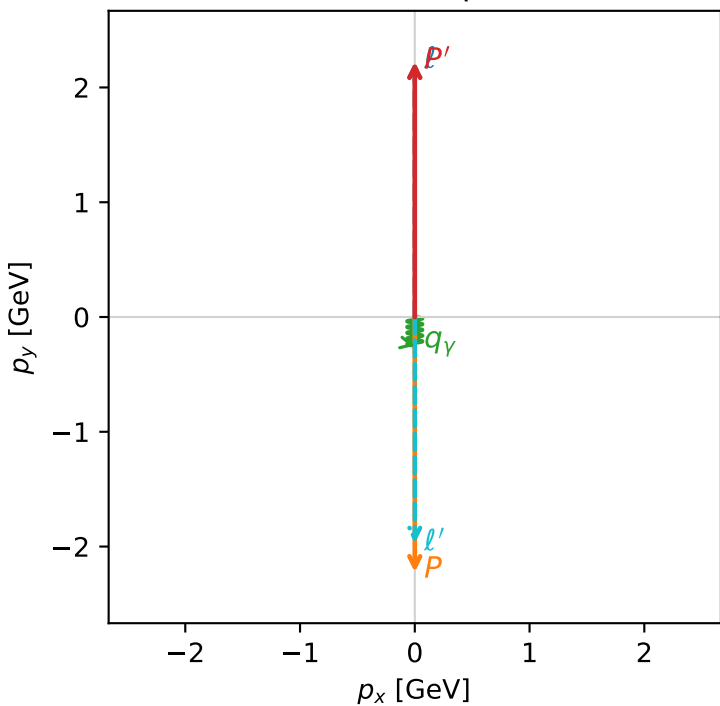


f3_low_egamma_txtx_region_1 F_3=0.999999
 region: low_Egamma_TxTx_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |\bar{T}_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.22295$
 $\theta_{in}=1.57079$, $\phi_l=1.5708$, $\phi_p=4.71239$
 $E_\gamma=0.25$, $\phi_\gamma=4.71239$
 $Q^2=17.5445$, $x_B=0.883304$, $t=-19.7674$, $W^2=3.19769$, $y=0.963029$
 $\theta(l', \gamma)=0$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= 0.0000$
 ℓ' $E= 1.9758$ $p_x= 0.0000$ $p_y= -1.9729$ $p_z= 0.0000$
 P' $E= 2.4127$ $p_x= 0.0000$ $p_y= 2.2229$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.0000$ $p_y= -0.2500$ $p_z= 0.0000$

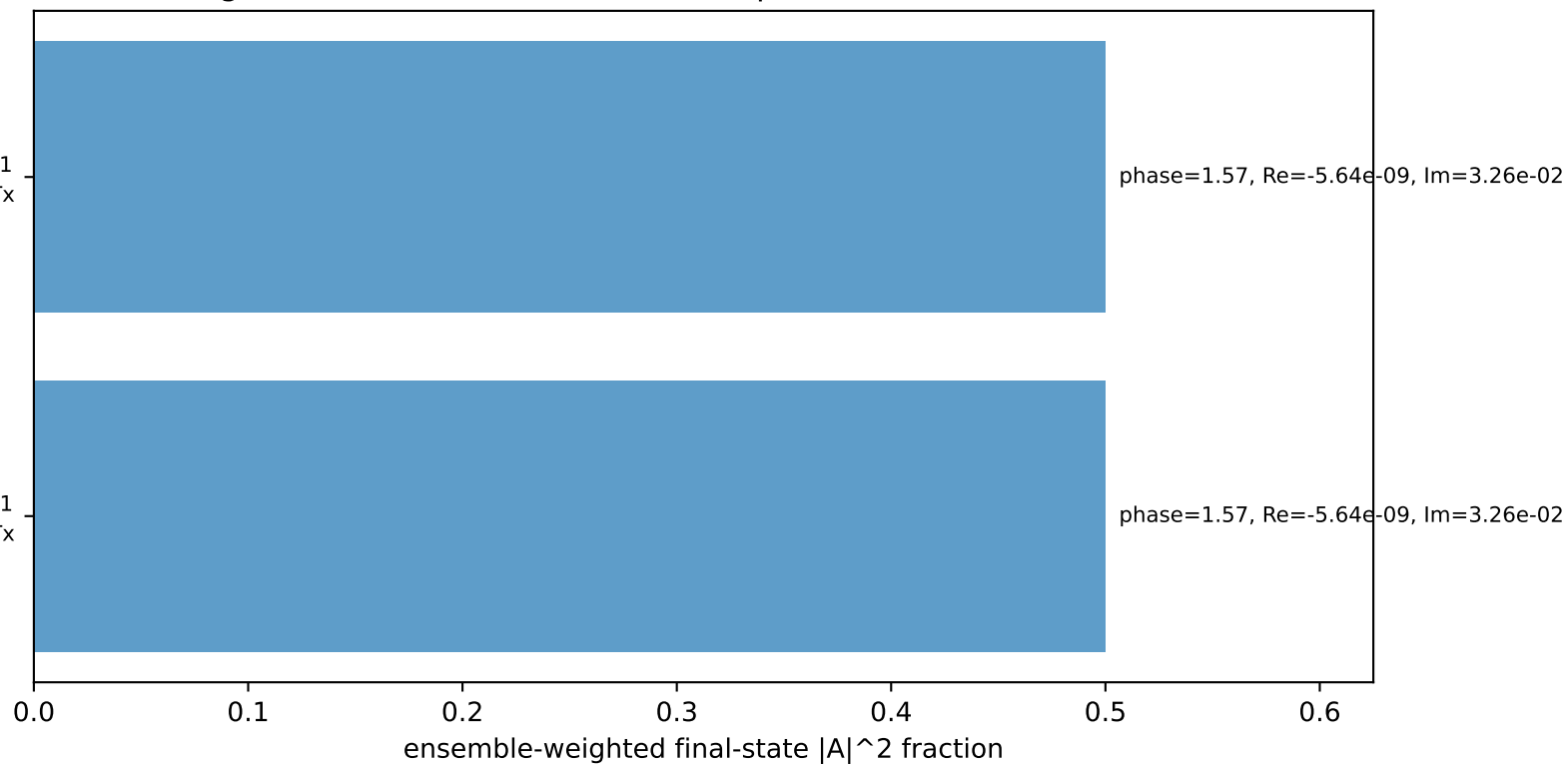
Transverse plane



$h_e=-1, h_p=+1, h_\gamma=+1$
 electron Tx, proton Tx

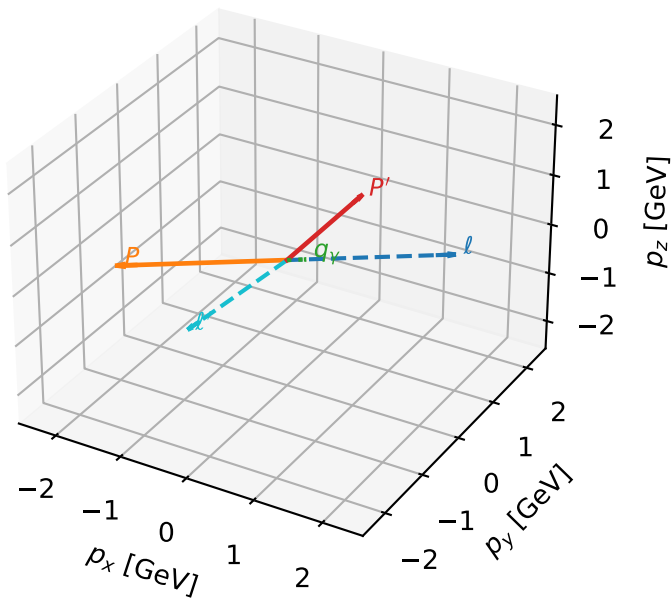
$h_e=+1, h_p=-1, h_\gamma=-1$
 electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



Tx muon + Tx proton characteristic max F_3 configuration

3D momenta

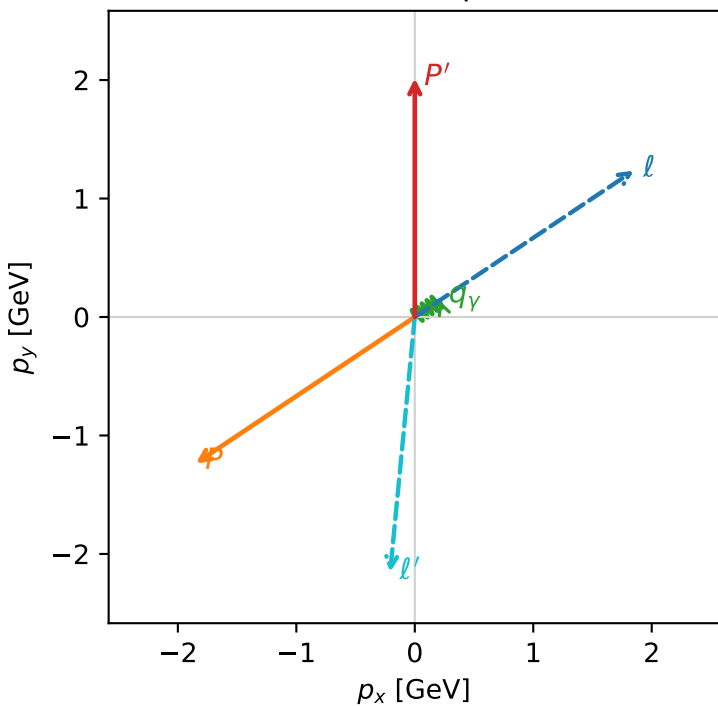


f3_low_egamma_txtx_region_2 F_3=0.907224
 region: low_Egamma_TxTx_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |\bar{T}_{x,p}\rangle$

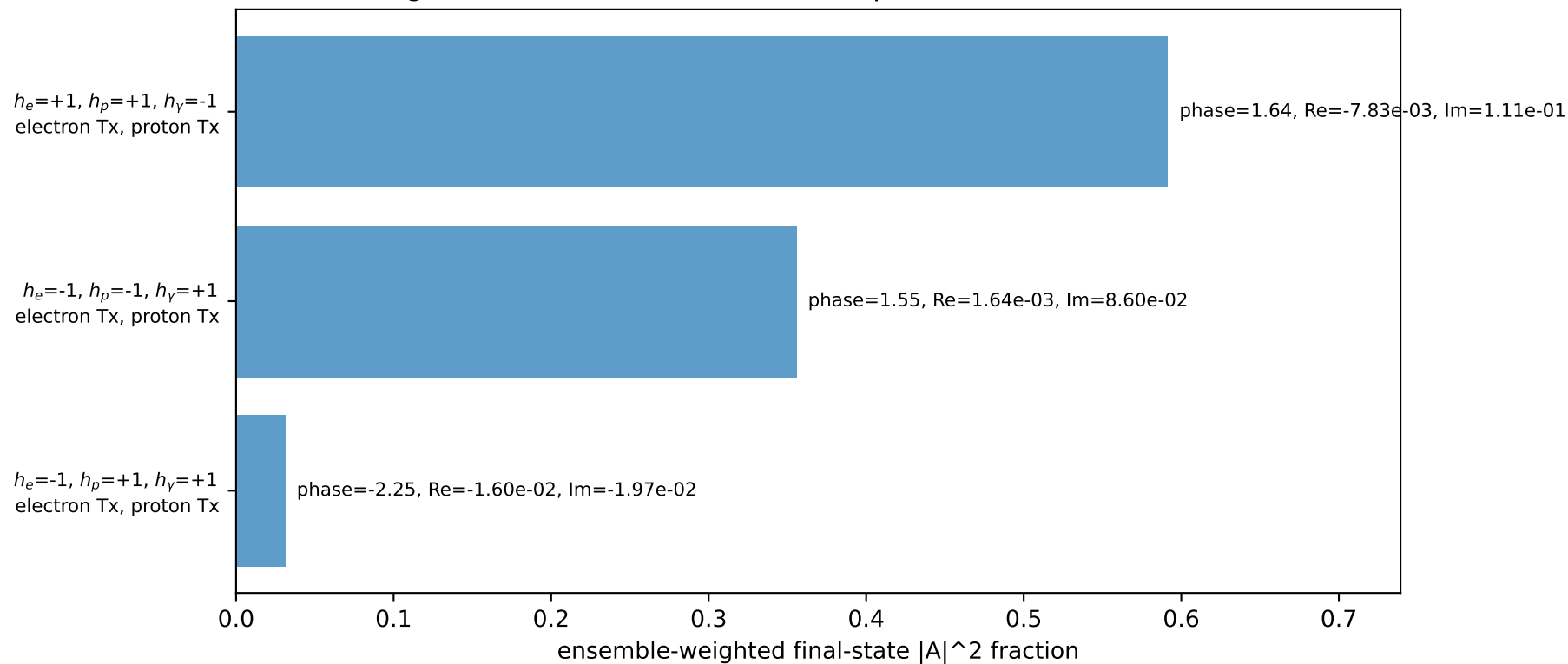
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.01469$
 $\theta_{in}=1.57079$, $\phi_l=0.589049$, $\phi_p=3.73064$
 $E_\gamma=0.25$, $\phi_\gamma=0.589049$
 $Q^2=15.708$, $x_B=0.96608$, $t=-13.9416$, $W^2=1.43137$, $y=0.788349$
 $\theta(l', \gamma)=129.263$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 1.8484$ $p_y= 1.2351$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.8484$ $p_y= -1.2351$ $p_z= 0.0000$
 l' $E= 2.1662$ $p_x= -0.2079$ $p_y= -2.1536$ $p_z= 0.0000$
 P' $E= 2.2223$ $p_x= 0.0000$ $p_y= 2.0147$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2079$ $p_y= 0.1389$ $p_z= 0.0000$

Transverse plane

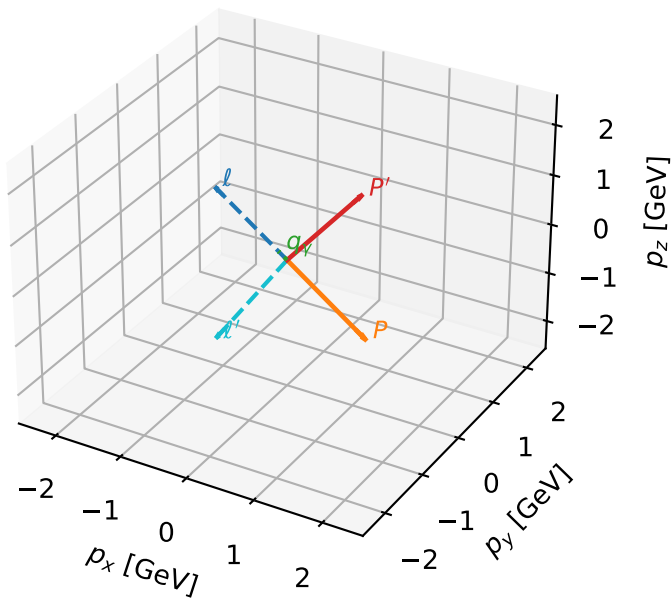


Leading initial-ensemble/final-state components (N=3, retained=97.8%)



Tx muon + Tx proton characteristic max F_3 configuration

3D momenta

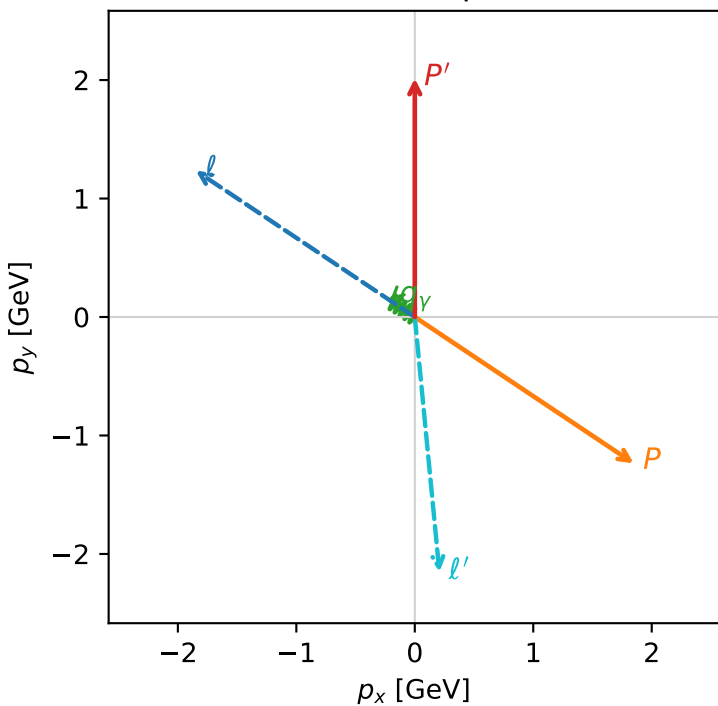


f3_low_egamma_txtx_region_3 F_3=0.907224
 region: low_Egamma_TxTx_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |\bar{T}_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.01469$
 $\theta_{in}=1.57079$, $\phi_l=2.55254$, $\phi_P=5.69414$
 $E_\gamma=0.25$, $\phi_\gamma=2.55254$
 $Q^2=15.708$, $x_B=0.96608$, $t=-13.9416$, $W^2=1.43137$, $y=0.788349$
 $\theta(l', \gamma)=129.263$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -1.8484$ $p_y= 1.2351$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.8484$ $p_y= -1.2351$ $p_z= 0.0000$
 l' $E= 2.1662$ $p_x= 0.2079$ $p_y= -2.1536$ $p_z= 0.0000$
 P' $E= 2.2223$ $p_x= 0.0000$ $p_y= 2.0147$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.2079$ $p_y= 0.1389$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=97.8%)

